

(vi) Measuring Asymmetry

- * Skewness
- * Symmetry
- * Tail weight

(vii) Continuous Distribution

- * Normal distribution
- * Probability density
- * Continuous variables.

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Unit-3 Solved Problems

① Descriptive Statistics

- Data summarization
- Mean, mode, median & Range

P.1 Marks of 10 students are

45, 52, 60, 48, 75, 60, 55, 68, 72, 50

$$\text{Mean} = \frac{\sum x}{n} = \frac{45 + 52 + 60 + 48 + 75 + 60 + 55 + 68 + 72 + 50}{10} = \frac{585}{10} = 58.5$$

Step 2:

$$\begin{aligned} \text{Median} &= 45, 48, 50, 52, 55, 60, 68, 72, 75 \\ &= \frac{50 + 52}{2} \\ &= 51 \end{aligned}$$

Steps: mode > 60

$$\text{Range} = 75 - 45 = 30$$

Unit 3 - solved problems

- ② The following are the number of hours 15 students spent studying.
2, 4, 3, 5, 2, 6, 4, 3, 5, 7, 2, 4, 6, 5, 3

Sol: given,

2, 4, 3, 5, 2, 6, 4, 3, 5, 7, 2, 4, 6, 5, 3

Step 1: Arrange the values

2, 3, 4, 5, 6, 7

Step 2: Count the frequency

Study Hours	Frequency
2	3
3	3
4	3
5	3
6	2
7	1
Total	15

Step 3: calculate Percentage

$$\text{Percentage} = \frac{f}{N} \times 100$$

f = Frequency

N = total number of observations = 15

Hours	Frequency	Percentage
2	3	$3/15 \times 100 = 20\%$
3	3	20%
4	3	20%
5	3	20%
6	2	$2/15 \times 100 = 13.33\%$
7	1	$1/15 \times 100 = 6.67\%$
Total	15	100%

Answer: The highest Frequency is 3, occurring for 2, 3, 4 and 5 hours.

Q1) consider the following dataset, 10, 12, 15, 18, 20, 22, 25, 28, 30, 35 Find q_1, q_2, q_3, IQR .

Sol: given,

10, 12, 15, 18, 20, 22, 25, 28, 30, 35

$$n=10$$

Find q_2

$n=10$, the median is the avg of the 5th & 6th values

$$q_2 = \frac{20+22}{2}$$

$$q_2 = 21$$

Find q_1

10, 12, 15, 18, 20

$$q_1 = 15 \text{ middle value}$$

Find q_3

upper half

22, 25, 28, 30, 35

The middle value is

$$q_3 = 28$$

calculate IQR

$$IQR = q_3 - q_1$$

$$IQR = 28 - 15$$

$$IQR = 13$$

$$\therefore q_1 = 15, q_2 = 21, q_3 = 28, IQR = 13$$

Q2) consider the dataset, identify outlier using the IQR method.

Sol: data: 10, 12, 13, 14, 15, 16, 17, 18, 20, 50

Find q_1

lower half:

10, 12, 13, 14, 15

$$q_1 = 13$$

Find q_3

upper half

16, 17, 18, 20, 50

$$q_3 = 18$$

IQR

$$IQR = q_3 - q_1$$

$$= 18 - 13$$

$$IQR = 5$$

calculate lower boundary

$$L = Q_1 - 1.5(IQR)$$

$$= 13 - 1.5(5)$$

$$= 13 - 7.5$$

$$L = 5.5$$

Identify the outlier

An value:

$$< 5.5 \text{ or } > 25.5$$

Here:

$$50 > 25.5$$

$\therefore 50$ is an outlier

Outlier treatment

$$\text{Median} = \frac{15 + 16}{2} = 15.5$$

$\therefore 50$ can be replaced by 15.5

$$\therefore \text{Outlier} = 50$$

calculate upper boundary

$$U = Q_3 + 1.5(IQR)$$

$$= 18 + 1.5(5)$$

$$= 18 + 7.5$$

$$U = 25.5$$

⑤ Suppose the exam scores are 40, 42, 45, 46, 48, 50, 52, 90. detect the outliers using 2-score method.

Sol:

given,

40, 42, 45, 46, 48, 50, 52, 90

given,

$$\mu = 51.625$$

$$\sigma = 15.69$$

Formula

$$Z = \frac{x - \mu}{\sigma}$$

$$\text{As } x = 90$$

$$Z = \frac{90 - 51.625}{15.69}$$

$$Z = \frac{38.375}{15.69}$$

$$Z \approx 2.45$$

Apply the rule

$$|Z| > 3$$

$$2.45 < 3$$

$\therefore 90$ is not an outlier

consider 2, 3, 3, 4, 4, 5, 5, 6, 10 determine the type of skewness

given 2, 3, 3, 4, 4, 5, 5, 6, 10

calculate mean:

$$\text{mean} = \frac{\sum x}{n}$$

$$\sum x = 2 + 3 + 3 + 4 + 4 + 5 + 5 + 6 + 10 = 42$$

$$n = 9$$

$$\text{mean} = \frac{42}{9}$$

$$\text{Mean} = 4.67$$

Median:

$$\frac{9+1}{2} = 5$$

so value:

$$\boxed{\text{median} = 4}$$

compare mean and median:

$$\text{Mean} = 4.67$$

$$\text{Median} = 4$$

$$\text{mean} > \text{median}$$

the distribution is positively skewed.

$\therefore$ Distribution is positively (right) skewed.

⊕ The waiting time for a bus is uniformly distributed between 0 and 20 minutes. Find the probability that a passenger waits between 5 and 12 minutes.

Sol:

To find:

$$P(5 < x < 12)$$

Formula:

$$P(c < x < d) = \frac{d-c}{b-a}$$

$$a=0, b=20, c=5, d=12$$

Substitute,

$$P(5 < x < 12) = \frac{12-5}{20-0}$$

$$= \frac{7}{20}$$

$$= 0.35$$

convert to Percentage:

$$0.35 \times 100 = 35\%$$

$$P(5 < X < 14) = 0.35$$

$$= 35\%$$

$$\boxed{35\%}$$

⑤ The heights of students are normally distributed with:

$\mu = 170\text{ cm}$, $\sigma = 10\text{ cm}$. Find the z-score for a student whose height is 185 cm.

Sol:

Heights are normally distributed with:

$$\mu = 170\text{ cm}$$

$$\sigma = 10\text{ cm}$$

and the z-score for $x = 185\text{ cm}$

Formula:

$$z = \frac{x - \mu}{\sigma}$$

Substitute

$$z = \frac{185 - 170}{10}$$

$$z = \frac{15}{10}$$

$$\boxed{z = 1.5}$$

Final Answer

$$\boxed{z = 1.5}$$

The student's height is 1.5 standard deviation above the mean.

Q using the previous example, what is the probability that a randomly selected student has a height less than 185cm:

sol:

Given sol:

$$z = 1.5$$

use the standard d-normal table

$$\text{for } z = 1.5$$

$$P(z < 1.5) = 0.9332$$

convert to Percentage

$$0.9332 \times 100 = 93.32\%$$

Final Answer:

$$P(X < 185) = 0.9332$$

$$\boxed{93.32\%}$$

Interpretation:

Approximately 93.32% of students are expected to have heights below 185 cm.